DAILY RESUSCITATION LEARNING | 13 NOVEMBER 2026

# RSI induction: etomidate versus ketamine

Physiologically difficult airway | New 2026 randomized-evidence synthesis | Emergency and transport practice

## THE PHYSIOLOGICAL PROBLEM

Induction removes sympathetic compensation, reduces venous return when positive-pressure ventilation begins, and can precipitate cardiovascular collapse in patients with shock, RV dysfunction, tamponade, severe hypoxaemia or metabolic acidosis. Ketamine has sympathomimetic effects but also direct myocardial depression; catecholamine-depleted patients may become hypotensive. Etomidate usually preserves immediate haemodynamics but inhibits adrenal steroidogenesis.

## THE NEW EVIDENCE — MCDOUGALL ET AL. (CHEST 2026)

Systematic review and network meta-analysis of **21 randomized trials, 6,031 critically ill patients** across ED, ICU, operating room and prehospital settings. Against etomidate, ketamine increased peri-intubation haemodynamic instability: **RR 1.31 (95% CI 1.15–1.48; moderate certainty)**. Ketamine–propofol appeared to reduce instability versus etomidate: RR 0.44 (0.27–0.72), but certainty was low. Ketamine versus etomidate for starting continuous post-intubation vasopressors: RR 0.80 (0.50–1.28; low certainty). Etomidate increased adrenal suppression. Outcomes and definitions differed among trials; indirect network comparisons are less secure.

Corroboration: Zampieri et al., *Critical Care* 2026 (9 RCTs, 4,672 patients) found similar short-term mortality for ketamine versus etomidate (OR 0.96, 95% CI 0.80–1.16) but more cardiovascular collapse with ketamine (OR 1.44, 1.20–1.71). These reviews share trials and must **not** be treated as independent patient populations.

## BEDSIDE PRE-INDUCTION SEQUENCE

**1. Recognize the phenotype.** Identify shock trajectory, oxygenation reserve, severe metabolic acidosis, RV failure, LV dysfunction, tamponade or obstructive physiology. Focused cardiac/lung POCUS may change the plan but must not delay rescue of a crashing airway.

**2. Optimize before paralysis where feasible.** Prepare suction, video laryngoscopy, bougie, rescue supraglottic airway and eFONA. Preoxygenate using NIV or high-flow oxygen as appropriate; maintain apnoeic oxygenation. Assign a pressor operator and have norepinephrine ready for a hypotension-prone patient. Correct severe hypovolaemia/haemorrhage while preparing the airway.

**3. Choose the induction agent for physiology.** Etomidate is a reasonable default when immediate haemodynamic stability dominates. Ketamine remains appropriate in selected patients, but do not assume it protects against collapse. Propofol can cause major vasodilation; ketofol network estimates are too uncertain to establish routine superiority. Individualize induction dose to illness severity and local protocols.

**4. Protect the first minutes after the tube.** Confirm with waveform capnography, assess bilateral ventilation and tube depth, avoid excessive airway pressures, titrate sedation and analgesia, reassess BP/ETCO2 and ultrasound when appropriate. Avoid abrupt loss of compensatory minute ventilation in profound metabolic acidosis.

## TRAPS / WHAT THE PAPER DOES NOT PROVE

A relative risk is not an absolute risk: baseline event rates vary by case mix and endpoint. The meta-analysis does **not** establish a mortality advantage of etomidate, a universal induction dose, or a safe option without preparation. Adrenal suppression is a biochemical signal, not proof that a single dose worsens patient-centred outcomes. Drug choice cannot compensate for inadequate resuscitation, poor preoxygenation or repeated airway attempts.

## ONE-MINUTE TAKEAWAY

**Ketamine is not reliably haemodynamically protective in critically ill RSI. Etomidate probably reduces immediate peri-intubation instability versus ketamine, but the best induction plan treats shock physiology, oxygenation and post-intubation haemodynamics as a single procedure.**

## SOURCES / REVIEW LEVEL

McDougall G et al. *Chest*. 2026. DOI: 10.1016/j.chest.2026.07.5239 (original abstract reviewed). Zampieri FG et al. *Critical Care*. 2026;30:296. DOI: 10.1186/s13054-026-06067-w (original abstract reviewed). Bedside sequence is synthesis, not a trial-tested protocol; use local Canadian airway/resuscitation policy.